$$|\beta| \leqslant 33.75°$$

$$\theta_1 = 90°$$

$$\theta_1' = \arcsin\left(\frac{2}{3}\right) = 41.81°$$

$$\beta_1 = 2\theta_1' - \theta_1 + \alpha_1 = 23.62° \qquad (7)$$

$$\beta_1 < \angle QOB \qquad (8)$$

$$i_C = \arcsin\left(\frac{1.400}{1.500}\right) = 68.96° \qquad (9)$$

$$\alpha' \leqslant 90° - 68.96° = 21.04° \qquad (10)$$

$$\sin\theta_2 = 2\sin\alpha_2 = 2\sin25° \qquad \theta_2 = 57.70°$$

$$\sin\theta_2' = \frac{1}{1.5}\sin\theta_2 \qquad \theta_2' = 34.30°$$

$$\beta_2 = 2\theta_2' - \theta_2 + \alpha_2 = 35.90° \qquad (11)$$

$$\sin\theta_3 = 2\sin\alpha_3 = 2\sin20° \qquad \theta_3 = 43.16°$$

$$\sin\theta_3' = \frac{1}{1.5}\sin\theta_3 \qquad \theta_3' = 27.13°$$

$$\beta_3 = 2\theta_3' - \theta_3 + \alpha_3 = 31.10° \qquad (12)$$

$$\alpha_3' = \theta_3 - \alpha_3 + \theta_3' = -3.97° \qquad (13)$$

